Quality Aware AI for Glaucoma Screening

Project Update 1 Database and Architecture

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We will test whether glaucoma prediction scores and image quality can define a gray zone for human review. The review rate will follow the fixed boundary, and the comparison model will select the same proportion of test images.

1 Train the glaucoma classifier

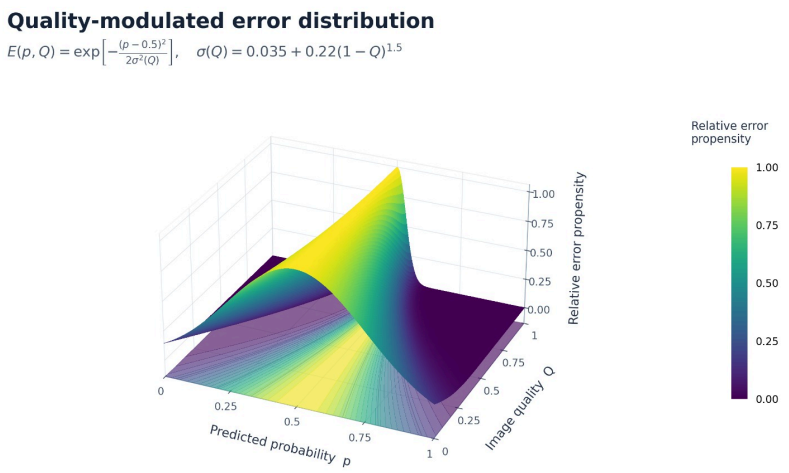
Use the glaucoma and normal fundus data from datasets ACRIMA, EyeAugmented, Kaggle Eye Diseases Classification, and ODIR-5K to train EfficientNet-B0. The model outputs a glaucoma prediction score. Keep ODIR paired eyes in the same data split, but retain each eye's own label for single-image training. The verified counts and inclusion rules are on the next page.

2 Train the image quality model

Train a separate model on EyeQ, which provides Good, Usable and Reject quality labels. Apply it to the same fundus images to obtain predicted quality scores. These scores are model predictions, not new human annotations. The quality model backbone has not yet been selected.

3 Map errors and define the review boundary

We first train a glaucoma classification model and an image quality assessment model. Each fundus image receives a glaucoma prediction score p and an image quality score q. We then compare the glaucoma predictions with the true labels in the validation set, identify misclassified images, and record their p and q values. Using these data, we plot the error distribution to see which ranges of prediction scores and image quality contain most of the model's errors.



Based on this distribution, we define a review boundary function using p and q to identify a gray zone for human review. This can be understood as placing a "cutting surface" within the error distribution and selecting the regions where errors are more concentrated for human review. When defining the boundary, we also check the distribution of correctly classified images in these regions to assess how many would also be selected for review.

4 Apply the fixed boundary

Once the boundary is defined, we fix it and apply it to an independent test set. For each new image, the system uses its glaucoma score p and quality score q directly to determine whether it falls within the gray zone. Images within this zone are sent for human review, while the remaining images keep the model's automatic classification results. The human-review rate is determined by the number of images selected by the boundary.

5 Compare models at the same observed review rate

To evaluate this method, we will choose a glaucoma binary classification model with good performance supported by reliable validation and compare both methods on the same test set. We first measure the proportion of images

our method selects for human review. We then have the comparison model select the same proportion of images for review based on prediction uncertainty. Finally, we compare the misclassification rates among the remaining images and separately count missed glaucoma cases. This allows us to assess the reliability of the automatic classification results retained by both methods at the same human-review rate.

Available Data

A Existing Project Data

The existing project collection includes glaucoma and normal fundus images from four sources, totaling 10,747 images under the original single-label criteria. ODIR inclusion is additionally defined using paired-eye groups.

A1 Image Counts by Source

Source	Glaucoma images	Normal fundus images
ACRIMA	396	309
EyeAugmented	2,330	2,615
Kaggle Eye Diseases Classification	1,007	1,074
ODIR-5K	200	2,816
Total	3,933	6,814

The first three sources use Glaucoma / Normal Retina categories. The ODIR row counts only images labeled exactly glaucoma or normal fundus.

A2 ODIR Paired Eyes and Training Labels

ODIR contains 7,000 images in 3,500 eye pairs. The current selection includes the following two mutually exclusive groups, totaling 1,124 pairs and 2,248 images.

Group	Selection rule	Pairs / images
Glaucoma	At least one eye labeled exactly glaucoma	123 / 246
Normal	Both eyes labeled exactly normal fundus	1,001 / 2,002
Total	Other pairs excluded from this group selection	1,124 / 2,248

The 246 images in glaucoma pairs are not all glaucoma-positive: the fellow eye may be normal or have another disease. Single-image training uses each eye's own label, and both eyes remain in the same training, validation or test split.

A3 Supplementary Images

Four usable glaucoma images were identified in Wide-Field / Kaggle-Eye_diseases / Glaucoma. Adding them would increase the original glaucoma image count to 3,937, while the normal fundus count would remain 6,814. These four images are currently listed separately as supplementary data.

B Additional External Datasets

We plan to incorporate the following external datasets to support glaucoma classification and image-quality assessment. They are listed separately and are not included in the existing project data totals.

B1 EyeQ

The Eye-Quality (EyeQ) Assessment Dataset is a reannotated subset of EyePACS for fundus image-quality assessment. It contains 28,792 retinal images assigned to three quality categories: Good, Usable and Reject. We plan to use EyeQ to train the quality-assessment component.

Source: https://github.com/HzFu/EyeQ

B2 Hillel Yaffe Glaucoma Dataset

The Hillel Yaffe Glaucoma Dataset (HYGD) contains 747 images, including 548 glaucoma images and 199 non-glaucoma images. We plan to use this dataset for fine-tuning. Image-quality scores calculated using FundusQ-Net range from 1 to 10; these are model-derived scores rather than manual quality annotations.

Source: https://physionet.org/content/hillel-yaffe-glaucoma-dataset/1.1.0/

B3 EyePACS-AIROGS-light-V2

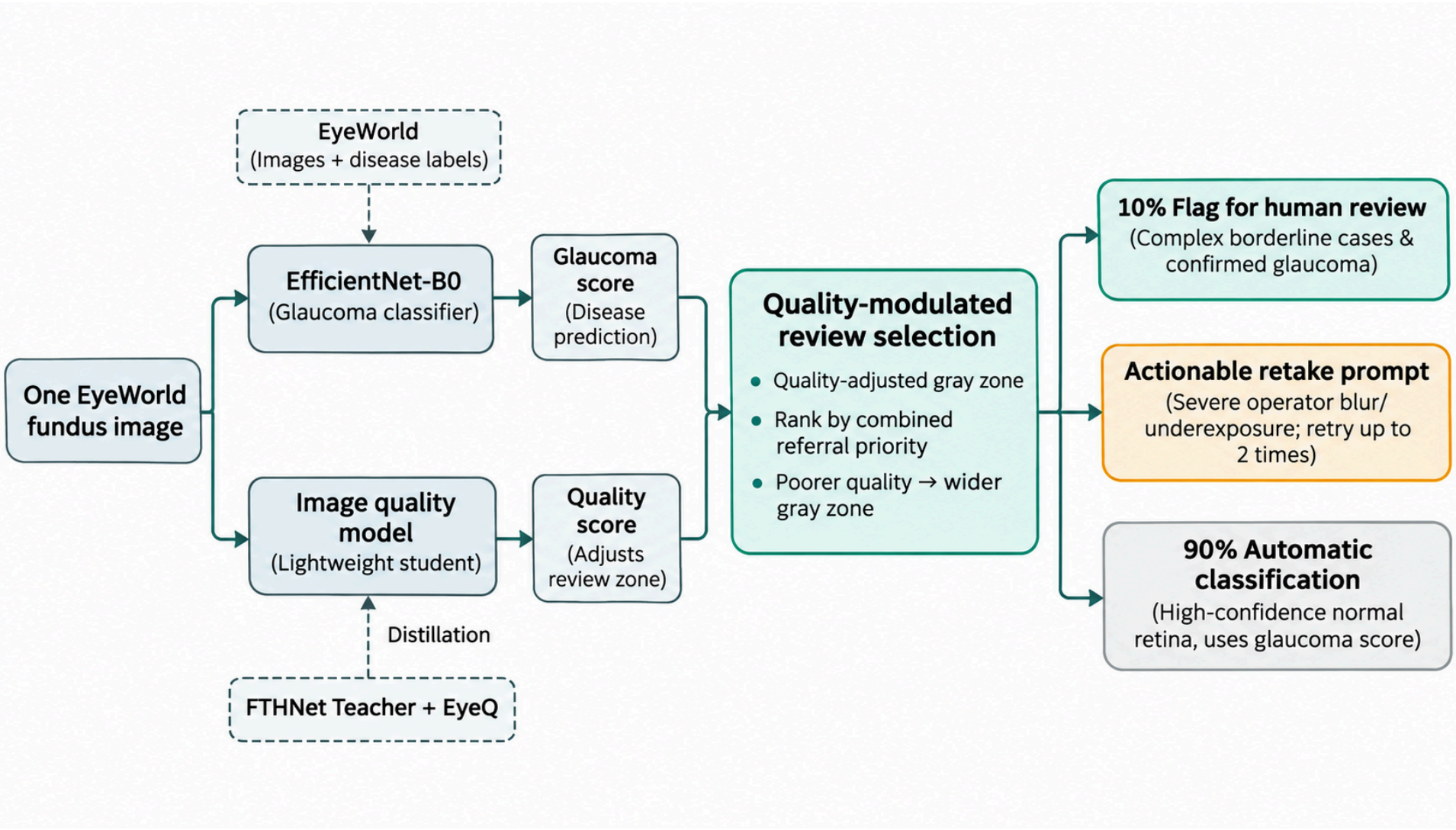
EyePACS-AIROGS-light-V2 provides predefined training, validation and test sets containing 4,000, 384 and 384 images, respectively. Each split includes two categories: referable glaucoma (RG) and non-referable glaucoma (NRG). We plan to use this dataset to train the glaucoma classifier.

Source: https://www.kaggle.com/datasets/deathtrooper/glaucoma-dataset-eyepacs-airogs-light-v2

B4 FQS Dataset and FTHNet

The FQS dataset provides continuous fundus image-quality scores and is presented alongside the FTHNet quality-assessment model. We plan to use FQS to train the model to predict a continuous quality score, complementing the three-category quality labels provided by EyeQ.

Source: https://pmc.ncbi.nlm.nih.gov/articles/PMC12627466/#Sec1


EyeWorld
(Images + disease labels)
One EyeWorld fundus image
EfficientNet-B0
(Glaucoma classifier)
Glaucoma score
(Disease prediction)
Image quality model
(Lightweight student)
Quality score
(Adjusts review zone)
Distillation
FTHNet Teacher + EyeQ
Quality-modulated review selection
Quality-adjusted gray zone
Rank by combined referral priority
Poorer quality → wider gray zone
10% Flag for human review
(Complex borderline cases & confirmed glaucoma)
Actionable retake prompt
(Severe operator blur/ underexposure; retry up to 2 times)
90% Automatic classification
(High-confidence normal retina, uses glaucoma score)